



# OPERATING SYSTEM DESIGN

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## Storage Management & System Security

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- Protection refers to a mechanism for **controlling the access of programs, processes, or users to the resources** defined by a computer system.
- Protection ensures that the resources of the computer are used in a consistent way.
- It ensure that each object accessed correctly and only by those processes that are allowed to do so.

### Goals of Protection

- We need to provide protection for several reasons.
- The most obvious is the need to prevent the **mischievous, intentional violation of an access restriction by user.**
- An unprotected resource cannot defend against use (or misuse) by an unauthorized or incompetent user. A protection-oriented system provides means to distinguish between **authorized and unauthorized usage.**

- Provide a mechanism for the enforcement of the policies governing resource use.
- These policies can be established in a variety of ways.
  - fixed in the design of the system
  - formulated by the management of a system
  - individual users
- A protection system must have the flexibility to enforce a variety of policies.

### Principles of protection

- The time-tested guiding principle for protection is **the Principle of least privilege**.
- It dictates that programs, users, and even systems be given enough privileges to perform their tasks.
- An operating system following the principle of least privilege implements its features, programs, system calls, and data structures so that failure or compromise of a component does the minimum damage to the system.
- The principle of least privilege can help produce a more secure computing environment.

### Domain of Protection

- A computer system is a collection of processes and objects.
- Object Refers to both **hardware objects** (such as the CPU, printer) and **software objects**(such as files, programs).
- Each object has a unique name that differentiates it from all other objects in the system, and each can be accessed only through well-defined and meaningful operations.

- A process should be allowed to access only those resources for which it has authorization.
- Furthermore, at any time, a process should be able to access only those resources that it currently requires to complete its task.
- A process operates within a **Protection Domain** that specifies the resources that the process may access.
- Each domain defines a set of objects and the types of operations that may be invoked on each object.
- The ability to execute an operation on an object is an **access right**.

- For example, if domain D has the access right  $\langle \text{file } F, \{\text{read}, \text{write}\} \rangle$ , then a process executing in domain D can both read and write file F; it cannot perform any other operation on that object.

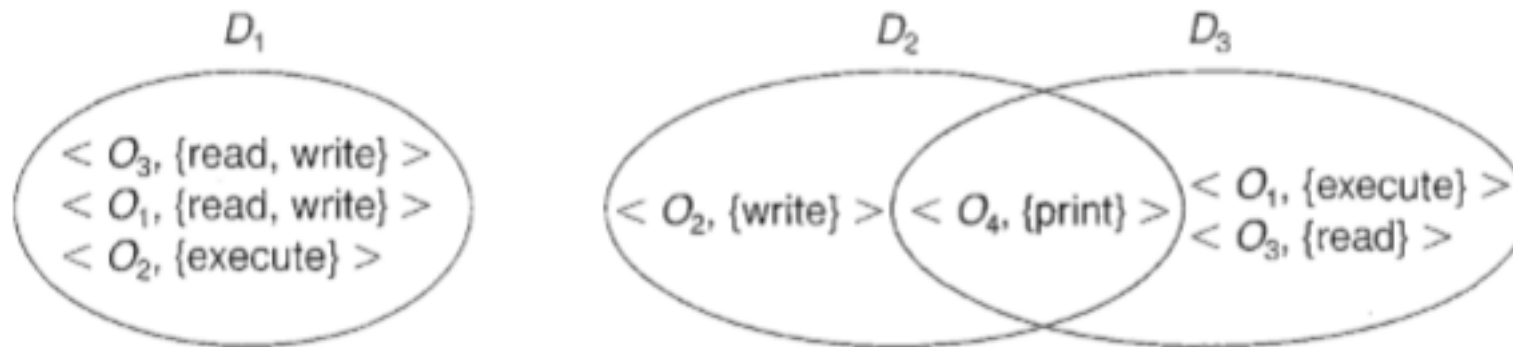


Figure 3.14 System with three protection domains.

A **domain** can be realized in a variety of ways:

- Each user may be a domain. In this case, the set of objects that can be accessed depends on the identity of the user.
- Each process may be a domain. In this case, the set of objects that can be accessed depends on the identity of the process.
- Each procedure may be a domain. In this case, the set of objects that can be accessed corresponds to the local variables defined within the procedure.



# THANK YOU

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